

Some Element properties

- *Element.id*
- *Element.attributes*
- *Element.className*
- *Element.innerHTML*
- *Element.outerHTML* (markup of the element including its content)

Element is the name of the element to be explored

https://www.w3schools.com/jsref/tryit.asp?filename=tryjsref_element_innerhtml_p

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other Element main properties

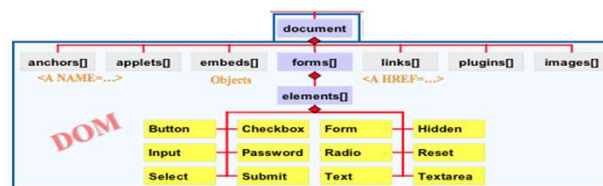
- *Element.style*
- *Element.setAttribute()*
- *Element.getAttribute()*
- *Element.addEventListener()*

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Which are the main DOM components?

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Document



“An HTML document”

Main properties

anchors[]
applets[]
embeds[]
forms[]
links[]
plugins[]

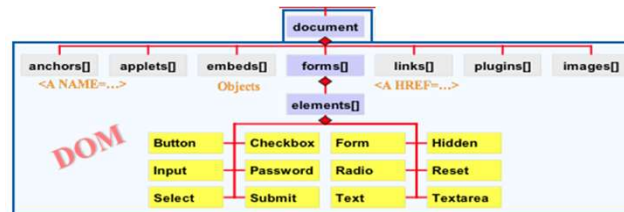
Other properties

lastModified
title, URL, referrer, cookie

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https://www.w3schools.com/jsref/tryit.asp?filename=tryjsref_document_lastmodified2

Document



"An HTML document"

Main methods

open()

close()

clear()

write()

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Examples

https://www.w3schools.com/jsref/tryit.asp?filename=tryjsref_doc_write5

https://www.w3schools.com/jsref/tryit.asp?filename=tryjsref_doc_open2

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DOM Element Selection

- How do you find and access HTML elements in a document?
- There are several ways traditionally used to find elements:
 - By id
 - By tag name
 - By class name
 - By name

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DOM Element Selection

- document.**getElementById**(*id*)

It returns (if found) the element that has the ID attribute with the specified value. Otherwise null.

Remember: only one element can have a specific id

https://www.w3schools.com/js/tryit.asp?filename=tryjs_dom_getelementbyid

- document.**getElementsByTagName**(*name*)

It returns (if found) an HTML collection containing all elements with the specified tag name (e.g. div, p, b and so on)

https://www.w3schools.com/js/tryit.asp?filename=tryjs_dom_getelementsbytagname2

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DOM Element Selection

- document.**getElementsByClassName**(*name*)

It returns (if found) an HTML collection of all HTML elements with the specified class name

https://www.w3schools.com/js/tryit.asp?filename=tryjs_dom_getelementsbyclassname

- document.**getElementsByName**(*name*)

It returns (if found) a NodeList containing all elements with the specified name

https://www.w3schools.com/jsref/tryit.asp?filename=tryjsref_doc_getelementsbyname_loop

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DOM Element Selection

- Today, to perform element selection is preferable to use:

- document.**querySelector**(*cssSelector*)

It returns (if found) the first element that matches the specified CSS selector. Otherwise null

- document.**querySelectorAll**(*cssSelector*)

It returns (if found) a NodeList of all elements that match the specified CSS selector. Otherwise an empty NodeList

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DOM Element Selection

```
const title = document.getElementById("title")
title.innerHTML = "Hello"
```

```
const title = document.querySelector("#title")
title.innerHTML = "Hello"
```

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DOM Structure modification

1. Create an element:

```
document.createElement(element)
```

2. Add an element to a page:

```
element.appendChild(element) or element.append(element)
```

↑
parent

↑
child

↑
parent

↑
child

3. Remove an element:

```
element.removeChild(element) or element.remove()
```

↑
parent

↑
child

↑
child

4. Replace an element:

```
element.replaceChild(newElement, oldElement)
```

or

38 `oldElement.replaceWith(newElement)`

DOM Structure modification

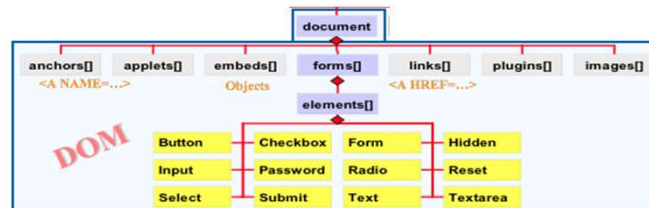
```

<!DOCTYPE html>
<html>
  <body>
    <div id="1">UNO</div>
    <div>TRE</div>
    <script>
      var node = document.createElement("div");
      //var textnode = document.createTextNode("DUE");
      //node.appendChild(textnode);
      node.innerText="DUE";
      document.getElementById("1").appendChild(node);
    </script>
  </body>
</html>

```

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Image



"An image embedded in an HTML document"

Main properties

border [width in pixels]

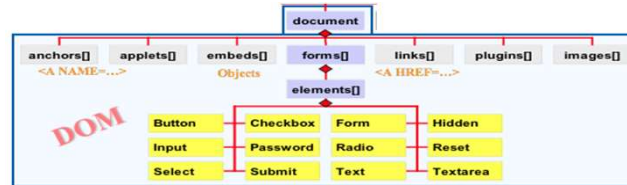
height

width

src [URL of the image to be displayed]

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Form



"An HTML input form"

Main properties

action *[destination URL]*

method *[get/post]*

name *[name of Form]*

target *[destination Window]*

Elements[] *[list of contained elements]*

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Main methods

reset()

submit()

Which are the main DOM events?

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Events

- Events are "things" that happen to HTML elements
- An HTML event can be something the browser does, or something a user does
- Examples:
 - The user selects, clicks, or hovers the cursor over a certain element
 - The user chooses a key on the keyboard
 - The user resizes or closes the browser window
 - A web page finishes loading
 - A form is submitted
 - An error occurs
- When an event happens a signal of some kind is generated. A mechanism by which an action can be automatically taken is provided

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Events

- The Event interface represents an event that takes place in the DOM
- There are many types of events, some of which use other interfaces based on the main Event interface. Event itself contains the properties and methods which are common to all events.

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Events

UIEvent:

- Input https://www.w3schools.com/jsref/obj_inputevent.asp
- Keyboard https://www.w3schools.com/jsref/obj_keyboardevent.asp
- Focus https://www.w3schools.com/jsref/obj_focusevent.asp
- Mouse https://www.w3schools.com/jsref/obj_mouseevent.asp
- Drag&drop https://www.w3schools.com/jsref/obj_dragevent.asp
- ...

Generic Events

- Events https://www.w3schools.com/jsref/obj_event.asp
- Animation https://www.w3schools.com/jsref/obj_animationevent.asp
- Clipboard https://www.w3schools.com/jsref/obj_clipboardevent.asp
- ...

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An event occurs...

- To react to an event, you attach an event handler to an element.
- An event handler is a piece JavaScript (a function) that runs when a specific event occurs
- We say we are registering an event handler when we define this code
- Two ways to play with DOM events :
 1. Event handler attributes: you define the handler directly in HTML using attributes (e.g. onclick). The attribute value is the JavaScript code you want to run when the event occurs
 2. Event listener: you attach an event listener to an element. The listener waits for the event to happen, and then calls the handler function.

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Example 1

- Event handler attribute

```
<!DOCTYPE html>
<html>
<head>
<script>
function displayDate() {
    document.getElementById("demo").innerHTML = Date();}
</script>
</head>

<body>
<p id="demo" onclick="displayDate()" >
Click me  to display the date</p>
</body>
</html>
```

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Example 2

- Event listener

```
<!DOCTYPE html>
<html>
<head>
    <title>Event Listener Example</title>
</head>
<body>
    <p id="demo">Click me to display the date</p>
    <script>
        function showDate() {
            document.getElementById("demo").innerHTML = Date();}

        document.getElementById("demo").addEventListener("click", showDate)
    </script>
</body>
</html>
```

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DOM in action: examples

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Events example 1

```
<body>
  <h1>Color Changer</h1>
  <br><br>
  Select Your Favorite Background Color:
  <form id="colorForm">
    <select id="color" onchange=setColor() >
      <option value="red">Red</option>
      <option value="green">Green</option>
      <option value="blue">Blue</option>
      <option value="white">White</option>
      <option value="yellow">Yellow</option>
      <option value="purple">Purple</option>
    </select>
  </form>
</body>
</html>
```



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Events example 1

```
<!DOCTYPE html>
<html>
<head>
  <title>Form Example</title>
  <script>
    function setColor() {
      var choice;
      choice = document.getElementById("color").selectedIndex;
      switch(choice) {
        case 0: document.body.style.backgroundColor = "#FF0000"; break;
        case 1: document.body.style.backgroundColor = "#00FF00"; break;
        case 2: document.body.style.backgroundColor = "#0000FF"; break;
        case 3: document.body.style.backgroundColor = "#FFFFFF"; break;
        case 4: document.body.style.backgroundColor = "#FFFF00"; break;
        case 5: document.body.style.backgroundColor = "#FF00FF"; break;
      }
    }
  </script>
</head>
```



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A more complex example A simple data entry validation page

PERSONAL INFORMATION

I campi contrassegnati con l'asterisco () sono obbligatori.*

* Cognome:
 Azienda:

Let's try!

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